Guinea Implementation Experience

Phase 1 Architecture, Governance, and Lessons Learned in Building a National Instant Payment Switch

PHASE 1 OVERVIEW

NimbaPay

Guinea will be launching its instant payment switch with two live use cases and a deliberately mixed roster of financial service providers in phase 1.

USE CASES

Person to Person (P2P)

Direct transfers between customer wallets and accounts across providers

Person to Merchant (P2M)

QR-based payments to merchants acquired by any participating DFSP

7 Digital Financial Service Providers (DFSPs) onboarded in Phase 1

PARTICIPATING INSTITUTIONS

3 Commercial Banks

3 Electronic Money Institutions

1 Microfinance Institution (MFI)

PROGRAM PARTNERS

The Partners Behind the National Switch

Guinea's instant payment system is a national program owned by the Banque Centrale de la République de Guinée and operated by GuiM as hub operator — delivered with the financial and institutional support of the partners below.

Gates Foundation

FUNDING · Financial support for the design and delivery of the national instant payment system.

AfricaNenda Foundation

PROGRAM MANAGEMENT · Program coordination, technical advisory, and stakeholder alignment.

PATH

FIDUCIARY SUPPORT · Fiduciary management of program funds and contracting of the system integrator.

Mojaloop Foundation

ECOSYSTEM STEWARDSHIP · Open-source platform, specifications, and the global implementer community.

IMPLEMENTATION · PROJECT MANAGEMENT

ThitsaWorks led system integration and technical delivery, and provided day-to-day project management — planning, DFSP onboarding coordination, and technical liaison with BCRG, GuiM, and the participating DFSPs.

Coordination: We Worked as One Team

DFSPs + **System Integrator** + **Project Financer** + **Local Project Leader, Regulator**

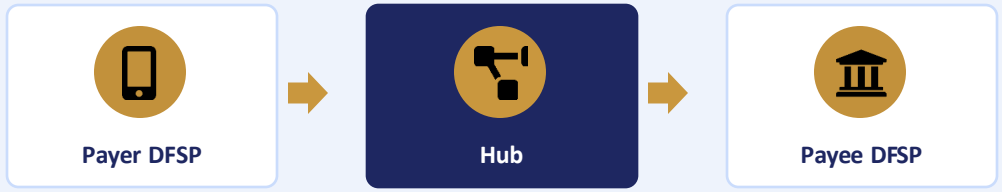
Weekly Working-Team Meetings	Regular cadence with the project working teams' representatives to unblock issues and keep delivery moving.
Weekly GuiM / AN Alignment	Standing sessions with GuiM and the AfricaNenda to align on shared priorities.
Shared Accountability Dashboards	Purpose-built dashboards tracked DFSP, SI accountability and collective alignment — shared with both the Steering Committee and the Regulator.
Cadence Escalation Near Go-Live	Meeting frequency increased and CEOs were brought in as launch approached — every stakeholder playing its role sustained the collective momentum.
Multiple Communication Channels	WhatsApp groups and other channels kept every stakeholder aligned in real time, beyond formal meetings.

ARCHITECTURE DECISIONS

Payee Routing & QR standard

Payee Discovery, No Central Registry

The payer DFSP includes the destination DFSP ID directly in the payment initiation call, so the switch routes instantly.



WHY

- Faster routing
- No central alias registry to build, govern, or maintain
- Account name still returned from the payee's own lookup response, so the payer can verify before confirming

EMVCo-Based QR Standard

MANDATORY DATA CARRIED IN THE QR STRING

QR generation

Generated by the merchant's acquiring institution by following the specification

Mandatory Data fields

DFSP ID, AccountID /Merchant ID, Terminal ID

Account Name Display

Account name is displayed from the account lookup response

WHY: No bottleneck for QR generation at any single party · No central QR registry to build or maintain

Transaction Fee Management: Calculation, Bearer & Settlement

Fee calculation

Single source of truth for fee logic at hub— fee amounts are calculated at scheme level based on scheme rule

Payer DFSP takes the fee values from the hub and displays to the customers.

Removes fee-logic build and test burden from every DFSP's own roadmap.

Fee Bearer Model

An Amount Type flag tells the engine which side to calculate the fee against.

P2P · Amount Type: SEND

Fee borne by the Payer, added on top of the send amount.

P2M · Amount Type: RECEIVE

Fee borne by the Payee (merchant), deducted from the receive amount.

Settlement (Current State)

01 Fee summary report generated at settlement time

02 Operations staff reconcile amounts owed per DFSP

03 Manual deposit / withdraw against payee DFSP

A deliberate Phase 1 trade-off — a natural candidate for future automation.

LESSONS LEARNED

The Biggest Challenge Was Engagement, Not Technology

Some DFSPs relied on third-party technical teams, introducing real-world coordination complexity beyond the technical scope of the project.

Multiple Time Zones

Vendor teams spread across regions slowed synchronous coordination and issue resolution.

Multiple Vendors

Each DFSP's third-party relationships added negotiation and dependency layers outside direct control.

Team Structure

Internal DFSP team composition and decision-making chains varied widely in maturity and speed.

TAKEAWAY

Assess team structure and availability — not just technical readiness — as a core criterion when selecting DFSPs for onboarding.

LESSONS LEARNED

Sharing Resources Ahead of Need

DFSPs not yet included in the current phase still received full access to specifications and integration resources — so they could prepare in advance.

PHASE 1 · ACTIVE

Onboarding & Go-Live

The core team fully specifies the switch, publishes documentation, and onboards the initial cohort of 7 DFSPs.

All specifications, sandbox access, and integration guides are published openly to the ecosystem — including future-phase DFSPs.

PHASE 2 · PREPARING

Early Self-Preparation

Future-phase DFSPs use the published resources to begin integration work well before their onboarding slot opens.

RESULT

Some DFSPs completed significant integration work before Phase 2 formally began — and the team's role shifted to answering targeted questions rather than driving onboarding from zero.

LESSONS LEARNED

An Ecosystem Advocate Changes the Trajectory

Having a trusted local organization able to advocate for technical decisions — to the regulator and to the ecosystem — proved to be one of the project's greatest assets.

GuiM

Local Ecosystem Advocate

Regulator communication

Conveyed and contextualized the project's technical decisions to the regulator on the team's behalf.

DFSP engagement push

Actively pushed DFSPs toward engagement, reinforcing timelines and expectations from within the local ecosystem.

Why It Mattered

- A credible local voice lends legitimacy to technical choices in front of the regulator
- Reduces the project team's own advocacy burden across many bilateral relationships
- Speeds up ecosystem-wide alignment on decisions that would otherwise take longer to land

Thank You